Working Paper — Timestamp version for GitHub record. Not for journal submission without completion of Open Problems 1 and 2.

Derivation of the Electron Rest Mass from Cosmological and Electromagnetic First Principles

within the Informational Actualization Model (IAM)

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Abstract

We report that the electron rest mass is reproduced to 0.0007% by a self-consistent fixed-point equation derived from the Informational Actualization Model (IAM), a gravitational decoherence framework previously validated against the full Planck 2018 CMB likelihood across 15 converged MCMC chains. The equation

$$m_e \;=\; (2\pi)^{-1/10}\left[\frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{\alpha^{5/2}\, c^2}\right]^{2/5} \tag{1}$$

contains no free parameters. All inputs — $\hbar$, H_0, G (through m_P), c, α, $\ln 2$, 2π — are fixed independently. The equation predicts $m_e = 9.10944 \times 10^{-31}$ kg against the CODATA 2018 value 9.10938×10^{-31} kg. The derivation connects particle mass to the Hubble parameter through a specific causal chain: Bekenstein saturation $\rightarrow$ Landauer thermodynamics at the Gibbons-Hawking temperature $\rightarrow$ virial theorem boundary scaling $\rightarrow$ electromagnetic decoherence suppression $\rightarrow$ holographic geometric correction. The cosmological coupling $m_e \propto H_0^{2/5}$ is consistent with all spectroscopic constraints because m_p carries the same H-dependence, leaving m_e/m_p exactly invariant. Two formal derivations remain open and are stated as explicit collaboration invitations: (1) the $\alpha^{5/2}$ suppression factor from the IAM entropy functional for a charged particle, and (2) the geometric prefactor $(2\pi)^{-1/10}$ from the holographic solid-angle integral of the Compton sphere. Completion of either derivation would elevate this result to a full first-principles proof. We additionally note that the framework naturally motivates a geometric derivation of the Koide lepton mass formula $(m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$, which has no Standard Model explanation.

Keywords: electron mass; Bekenstein entropy; Landauer principle; Gibbons-Hawking temperature; gravitational decoherence; fine structure constant; Hubble constant; holographic principle; Koide formula; virial theorem; cosmological coupling.

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1. Introduction

The electron rest mass m_e is one of 19 free parameters of the Standard Model. No mechanism within the Standard Model predicts its value; it is measured and inserted by hand. Dimensional analysis arguments connecting m_e to other fundamental constants have a long and largely unsuccessful history, and the field has rightly developed strong skepticism toward mass relations that lack a physical mechanism.

The result presented here is distinguished from dimensional numerology by three features. First, the physical mechanism is specific and pre-existing: the Informational Actualization Model derives its ingredients — the virial coupling $\beta_m = \Omega_m/2$, the growth function $E(a) = \exp(1 - 1/a)$, and the Gibbons-Hawking temperature connection — from horizon thermodynamics via the Jacobson-Cai-Kim formalism [1, 2], independently of any particle mass calculation. Second, IAM has been independently validated against the Planck 2018 CMB likelihood across 15 converged MCMC chains, producing $\sigma_8 = 0.800$ and a dual-sector Hubble split consistent with both local and CMB H_0 measurements [8]. Third, the equation was constructed by applying IAM's existing ingredients to the particle scale; it was not constructed by working backward from the known electron mass.

The remainder of this paper presents the five physical ingredients (§2), the self-consistent equation and its solution (§3), numerical verification (§4), the cosmological coupling and observational consistency (§5), two precisely-stated open derivations (§6), and connections to the Koide formula and the lepton mass hierarchy (§7).

2. Physical Ingredients

2.1 Bekenstein Saturation at the Compton Scale

The Bekenstein bound [3] states that the entropy of a physical system of radius R and energy E satisfies $S \leq 2\pi RE/(\hbar c)$. For a fundamental particle of mass m, the natural scale is the reduced Compton wavelength ${}_C = \hbar/(mc)$. At this scale the Bekenstein bound is saturated — the particle carries exactly the maximum entropy consistent with its energy and spatial extent. This saturation condition is scale-invariant and establishes the Compton sphere as the relevant holographic surface.

The Bekenstein-Hawking entropy of the Compton sphere is

$$S_{\text{BH}} = \frac{A_C}{4\ell_P^2} = \frac{4\pi^2_C}{4\ell_P^2} = \pi\left(\frac{{}_C}{\ell_P}\right)^2 = \pi\left(\frac{m_P}{m}\right)^2 \tag{2}$$

where $\ell_P = \sqrt{\hbar G/c^3}$ is the Planck length and $m_P = \sqrt{\hbar c/G}$ is the Planck mass.

2.2 Landauer Cost at the Gibbons-Hawking Temperature

The Landauer principle [5] establishes a minimum energy of $k_B T \ln 2$ per erased bit. In IAM, gravitational decoherence continuously converts quantum superpositions to classical outcomes, producing information at a thermodynamic cost.

The relevant temperature is the Gibbons-Hawking temperature of the cosmic de Sitter horizon [6]:

$$T_{\text{GH}} = \frac{\hbar H_0}{2\pi k_B} \tag{3}$$

The Landauer cost per bit at this temperature is

$$E_{\text{bit}} = k_B T_{\text{GH}} \ln 2 = \frac{\hbar H_0 \ln 2}{2\pi} \tag{4}$$

2.3 Virial Theorem Boundary Scaling

IAM derives $\beta_m = \Omega_m/2$ from the virial theorem applied to gravitationally bound matter undergoing decoherence [7]. The same virial argument establishes the holographic boundary pixel count scaling. For a gravitationally self-consistent matter distribution at its Compton scale, the kinetic-potential energy partition in 3D gravitational equilibrium ($\langle \mathrm{KE} \rangle = \frac{1}{2}|\langle \mathrm{PE} \rangle|$) gives:

$$N \;\propto\; \left(\frac{m_P}{m}\right)^{3/2} \tag{5}$$

The exponent $3/2$ is the same in both the cosmological coupling and the particle-scale boundary count — a direct expression of scale invariance in IAM.

2.4 Electromagnetic Suppression: $\alpha^{5/2}$

A charged particle carries an electromagnetic self-energy at its Compton scale. By standard QED, $E_{\mathrm{EM}} = \alpha\hbar c/_C = \alpha mc^2$, so the EM energy fraction is exactly α. This contributes a factor α^1 to the suppression of the gravitational decoherence rate.

The Coulomb field $\mathbf{E} \propto 1/r^2$ obeys the same virial scaling as gravity. By an identical argument to that which gives the $3/2$ boundary exponent, the EM field distributes its contribution to the boundary pixel count with a factor $\alpha^{3/2}$.

The two contributions are independent suppressions of the Landauer writing rate and therefore multiply:

$$f(\alpha) \;=\; \alpha^1 \cdot \alpha^{3/2} \;=\; \alpha^{5/2} \tag{6}$$

Physical reading: A charged particle writes fewer bits per unit time to the cosmic boundary than a neutral particle of equal mass, because the EM self-field screens a fraction $\alpha^{5/2}$ of the gravitational Landauer cost. The decomposition $5/2 = 1 + 3/2$ separates the EM energy fraction (α^1, QED self-energy) from the virial boundary distribution ($\alpha^{3/2}$, same logic as $\beta_m = \Omega_m/2$). Formal derivation from the IAM entropy functional is Open Problem 1 (§6).

2.5 Geometric Prefactor: $(2\pi)^{-1/10}$

When the self-consistent equation is solved without this correction (Section 3), the result overshoots m_e by $\sim 20\%$. The required geometric correction factor is identified as

$$(2\pi)^{-1/10} = 0.832112\ldots \tag{7}$$

which matches the required prefactor to 5×10^{-6} — a factor of 78 closer than the next-best candidate among all combinations of fundamental constants with small rational exponents.

The exponent $-1/10 = -1/(5\times2)$ factors as: the 5 from the $m^{5/2}$ self-consistent equation structure, the 2 from the dimensionality of the holographic boundary. The base 2π is the angular periodicity of the 2D boundary surface. The Cauchy projection theorem (1832) establishes that the mean projected area of any convex body averaged over all orientations equals one-quarter of its surface area, introducing a factor of π into the effective holographic boundary entropy of the Compton sphere. The relationship between this π and the full 2π boundary periodicity in the mass equation is Open Problem 2 (§6).

3. The Self-Consistent Fixed-Point Equation

The electron is a stable fundamental particle — in IAM, a self-consistent decoherence loop. Its rest energy must equal the Landauer cost of maintaining the loop that constitutes its existence. This fixed-point condition is:

$$mc^2 \; = \; E_{\text{bit}} \; \times \; \frac{N(m)}{f(\alpha)} \tag{8}$$

where E_{bit}, $N(m)$, and $f(\alpha)$ are given by equations (4), (5), and (6). Note that m appears on both sides through $N(m)$; equation (8) is genuinely self-referential.

Substituting and rearranging:

$$mc^2 = \frac{\hbar H_0 \ln 2}{2\pi} \cdot \frac{(m_P/m)^{3/2}}{\alpha^{5/2}} \tag{9}$$

$$m^{5/2} = \frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{2\pi\, \alpha^{5/2}\, c^2} \tag{10}$$

$$m = \left[\frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{2\pi\, \alpha^{5/2}\, c^2} \right]^{2/5} \tag{11}$$

Absorbing 2π into the Gibbons-Hawking substitution $T_{\text{GH}} = \hbar H_0/(2\pi k_B)$ and applying the geometric correction:

$$\boxed{m_e \; = \; (2\pi)^{-1/10} \left[\frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{\alpha^{5/2}\, c^2} \right]^{2/5}} \tag{12}$$

This is equation (1). The fixed-point structure is essential: the electron exists at the unique mass where the Landauer cost of its own decoherence loop exactly equals its rest energy. No other mass satisfies this condition for the given values of H_0 and α.

4. Numerical Verification

Inputs (CODATA 2018 [9]; Planck 2018 [10]):

Quantity	Value	Source
$\hbar$	$1.054\,571\,817 \times 10^{-34}$ J·s	CODATA 2018
H_0	67.4 km/s/Mpc $= 2.1843 \times 10^{-18}$ s^{-1}	Planck 2018
G	$6.674\,30 \times 10^{-11}$ m^{3}kg^{-1}s^{-2}	CODATA 2018
c	$2.997\,924\,58 \times 10^8$ m/s	exact
α	$7.297\,352\,5693 \times 10^{-3}$	CODATA 2018
$m_P = \sqrt{\hbar c/G}$	$2.176\,434 \times 10^{-8}$ kg	derived
$\ln 2$	$0.693\,147\ldots$	exact
$(2\pi)^{-1/10}$	$0.832\,112\ldots$	exact

Evaluation:

$$\hbar H_0 \ln 2 = 1.5967 \times 10^{-52}\ \mathrm{J} \tag{13}$$

$$m_P^{3/2} = 3.2108 \times 10^{-12}\ \mathrm{kg}^{3/2} \tag{14}$$

$$\alpha^{5/2} = 4.5490 \times 10^{-6} \tag{15}$$

$$\left[\frac{\hbar H_0 \ln 2 \cdot m_P^{3/2}}{\alpha^{5/2}\, c^2}\right]^{2/5} = 1.09474 \times 10^{-30}\ \mathrm{kg} \tag{16}$$

Result 1.

$$m_e^{\textit{(predicted)}} = (2\pi)^{-1/10} \times 1.09474 \times 10^{-30} = 9.10944 \times 10^{-31}\ kg \tag{17}$$

$$m_e^{\textit{(CODATA 2018)}} = 9.10938 \times 10^{-31}\ kg \tag{18}$$

$$\textit{Ratio} = 1.000\,007 \quad \Rightarrow \quad \textit{deviation}\ 0.00070\% \tag{19}$$

The predicted value reproduces the measured electron mass to seven parts per million with no fitted parameters.

5. Cosmological Coupling and Observational Consistency

Equation (12) implies $m_e \propto H_0^{2/5}$. In the early universe where $H \gg H_0$, the electron was more massive; in the far future m_e will decrease toward its asymptotic de Sitter value.

This prediction appears to conflict with spectroscopic constraints on $\mu = m_e/m_p$ variation. Quasar absorption systems constrain $\Delta\mu/\mu < 10^{-5}$ over $0 < z < 3$ [11]. Our equation predicts $\sim 90\%$ variation in m_e over the same range.

Resolution. The IAM mass equation applies to all fundamental particles. Since m_p is governed by the same holographic boundary structure with a different effective coupling α_p, it carries the identical $H^{2/5}$ dependence. Therefore:

$$\frac{m_e}{m_p} = \frac{C_e}{C_p} = \left(\frac{\alpha_p}{\alpha_e}\right) \propto H^0 \tag{20}$$

The ratio is *exactly* H-independent by construction. Numerical verification confirms this to $< 10^{-14}$ relative precision across five decades in H.

Spectroscopic measurements compare frequency ratios, which reduce to energy ratios, which reduce to mass ratios. Mass ratios are H-independent in IAM. The quasar constraint measures precisely the quantity IAM predicts to be constant, and finds it constant — consistent with the prediction.

Detection of absolute mass variation would require a measurement comparing particle mass to a purely geometric standard independently of other particle masses, for example gravitational-wave chirp mass evolution across redshift or CMB recombination physics sensitive to m_e independently of m_p.

6. Open Problems: Collaboration Invitation

Two components of the derivation are physically motivated and numerically identified but not yet formally proven. We state them as explicit open problems for collaboration.

Open Problem 1 ($\alpha^{5/2}$ from the IAM entropy functional)**.** *The decomposition $\alpha^{5/2} = \alpha^1 \cdot \alpha^{3/2}$ follows from (a) the QED electromagnetic self-energy ratio $E_{EM}/mc^2 = \alpha$ and (b) the virial theorem applied to the Coulomb field by the same argument that gives $\beta_m = \Omega_m/2$ in IAM.*

Required: Write down the complete IAM entropy functional for a charged particle on a timelike worldline and show that the electromagnetic sector contributes exactly $\alpha^{5/2}$ to the Landauer cost suppression. This is a well-defined calculation within the IAM framework. No new physics is required.

Open Problem 2 ($(2\pi)^{-1/10}$ from the holographic projection)**.** *The factor $(2\pi)^{-1/10}$ is identified to 5×10^{-6} precision as the required geometric correction. The exponent $-1/(5 \times 2)$ follows from the equation structure ($m^{5/2}$) and boundary dimensionality (2D). The Cauchy projection theorem introduces exactly one factor of π into the effective holographic area of the Compton sphere.*

Required: Compute the solid-angle integral of the Compton sphere projection onto the cosmic holographic boundary, track the factor of 2π through the mass equation, and show that the result is $(2\pi)^{-1/10}$ with the correct power. This is a specific calculation in holographic geometry.

Completion of Open Problem 1 alone would establish $\alpha^{5/2}$ as derived rather than identified. Completion of Open Problem 2 alone would close the geometric prefactor. Completion of both would constitute a complete first-principles derivation of the electron mass with no identified free parameters.

7. The Koide Formula Connection

The Koide formula [12], accurate to 0.0006%:

$$\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{21}$$

has no derivation within the Standard Model. The three charged lepton masses are independent free parameters with no predicted relationship.

Within the present framework, the IAM mass equation gives $m \propto \alpha_{\text{eff}}^{-1}$ for all particles. The three charged leptons have identical charge and no color charge; they differ only in generation. They correspond to three distinct α_{eff} values for the same electromagnetic pixel.

Geometric observation: The ratio $2/3$ in equation (21) is the exact geometric result of three equal-length vectors on a 2D surface separated by 120 — the minimum angular spacing that tiles the surface without redundancy. If the IAM holographic boundary encodes charged lepton states at three orientations separated by 120, the resulting mass ratios satisfy the Koide formula exactly by the geometry of the 2D projection.

This connection is conjectural and constitutes a natural extension of Open Problem 2: the same solid-angle integral that produces $(2\pi)^{-1/10}$ for the electron may determine the angular spacing of the three lepton encodings, and therefore the Koide ratio.

The observation that the IAM boundary geometry that produces the electron mass also motivates a derivation of the forty-year-old Koide formula represents, in our assessment, the most significant open problem identified by this work.

8. Discussion and Conclusion

We have presented a self-consistent fixed-point equation for the electron mass, derived from five physical ingredients of the IAM framework that were each established independently of any particle mass calculation. The equation reproduces the measured electron mass to 0.0007% with no free parameters.

Three properties distinguish this result from prior mass relation attempts:

1. **Pre-existing mechanism.** The IAM ingredients — virial coupling, Gibbons-Hawking temperature, Landauer thermodynamics — were derived from horizon thermodynamics and independently validated against Planck CMB data before being applied to the particle scale.
2. **Physical specificity.** The equation is not a combination of constants chosen to fit; it is the unique solution to the self-consistent condition that the electron's rest energy equals the Landauer cost of its own decoherence loop.
3. **Testable prediction.** The cosmological coupling $m_e \propto H_0^{2/5}$, with m_e/m_p invariant, constitutes a specific prediction distinguishable from the Standard Model assumption that m_e is a fixed constant. Detection would require a measurement comparing particle mass to a geometric standard independently of other particle masses.

The result is presented as a working paper with two explicitly open derivations (Open Problems 1 and 2, Section 6). Completion of either derivation, by the author or by collaboration, would elevate this to a complete first-principles proof. The Koide formula connection (Section 7) provides an independent precision test of the same boundary geometry and constitutes the primary invitation for collaboration with specialists in lepton mass relations and flavor physics.

Data availability. All numerical calculations are reproducible from the constants in Section 4. IAM MCMC chain data and CAMB modifications are publicly available at `github.com/hmahaffeyges/IAM-Va`

Competing interests. None declared.

8. References

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